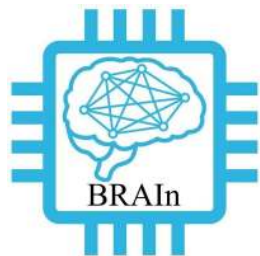




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Predicting Generalization in Few-shot Problems

Yassir Bendou , Vincent Gripon , Bastien Pasdeloup , Giulia Lioi

Introduction to Few-shot learning

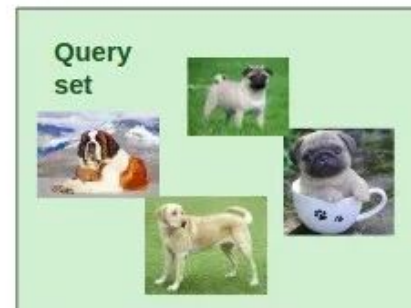
Some terminology...

In a Few-shot problem, we have:

- Nb of classes : **N-ways**.
- Support set : **K-shot** images par classe.

Typically:

5-ways | 1-shots or 5-shots

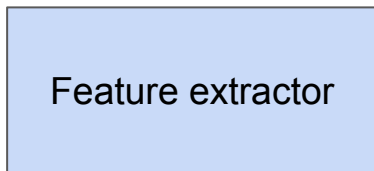


One few-shot problem

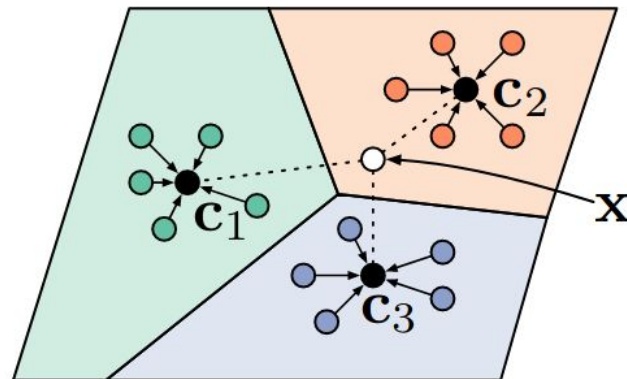
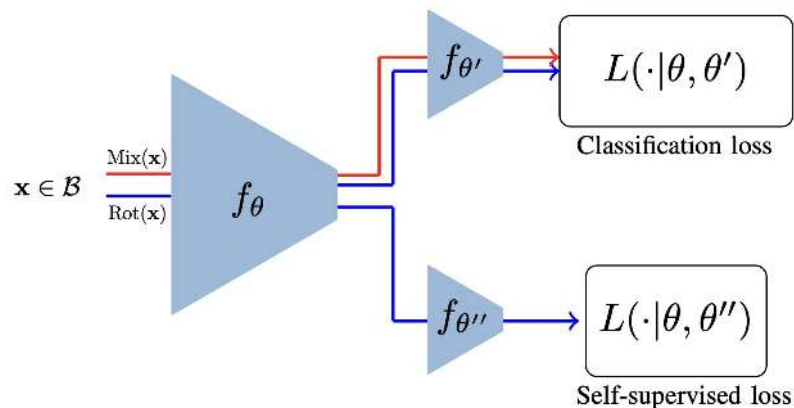
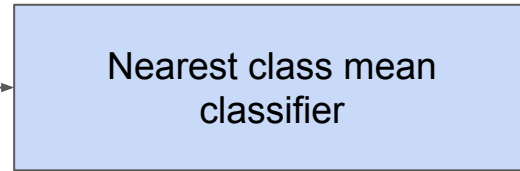
Introduction to Few-shot learning

In practice, we use a generic dataset

A generic dataset



A few-shot problem



Performance in Few-shot classification

Performance on 10k Few-Shot problems
sampled from the test dataset.

Base Dataset

64 classes

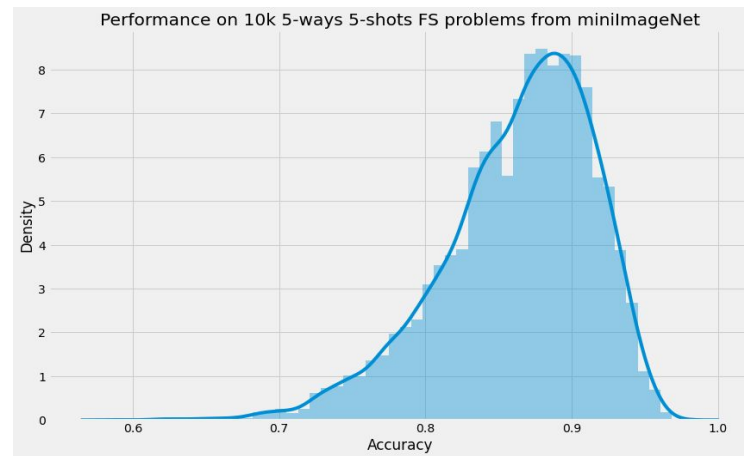
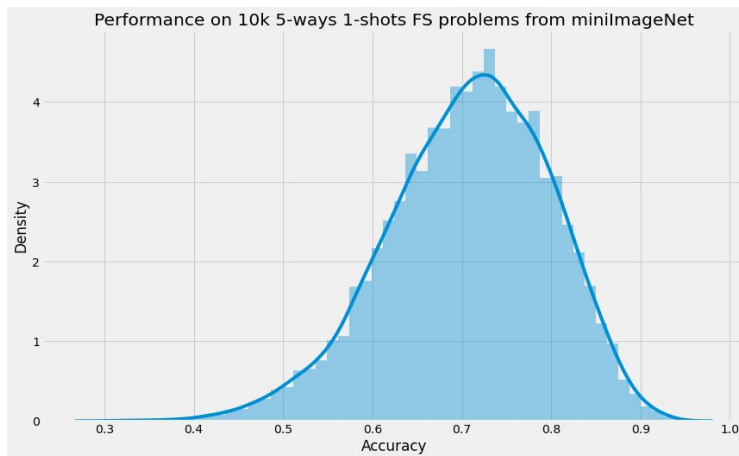
Validation Dataset

16 classes

Test Dataset

20 classes

Dataset	1-shot 5-ways	5-shots 5-ways
MinilImageNet	70.63% $\pm$ 0.20%	86.28% $\pm$ 0.12%
TieredImageNet	74.31% $\pm$ 0.22%	87.36% $\pm$ 0.15%



Predicting Generalization in Few-shot Problems

Problematics:

- Predicting performance of few-shot classifiers is hardly feasible with few training samples.
- Can we find better alternatives than the one-out cross-validation?

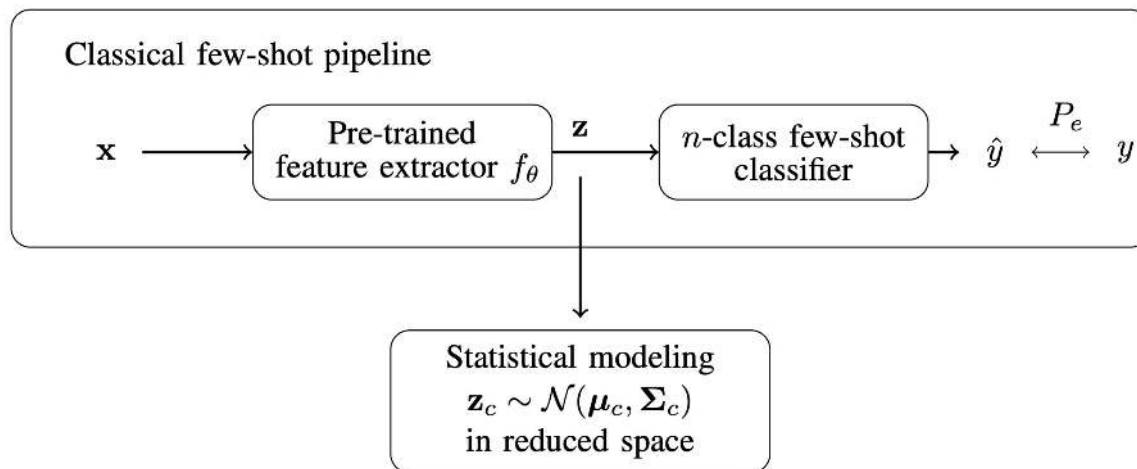
Setting:

- A few-shot task: N classes/ways with <20 shots per class.
- A pre-trained backbone.
- A distance-based classifier (Nearest Class Mean)

Goal: Predict the error of the classifier on the few-shot task.

Our approach

- Due to few available samples, we use a simple mathematical model with few parameters in order to infer the accuracy.
- We assume that the feature space is a multivariate gaussian distribution.



Some definitions

We assume that each class follows a gaussian distribution with a uniform prior across classes. The class-conditional densities are:

$$p(\mathbf{z} \mid y = c) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$$

The probability error can be defined as follow:

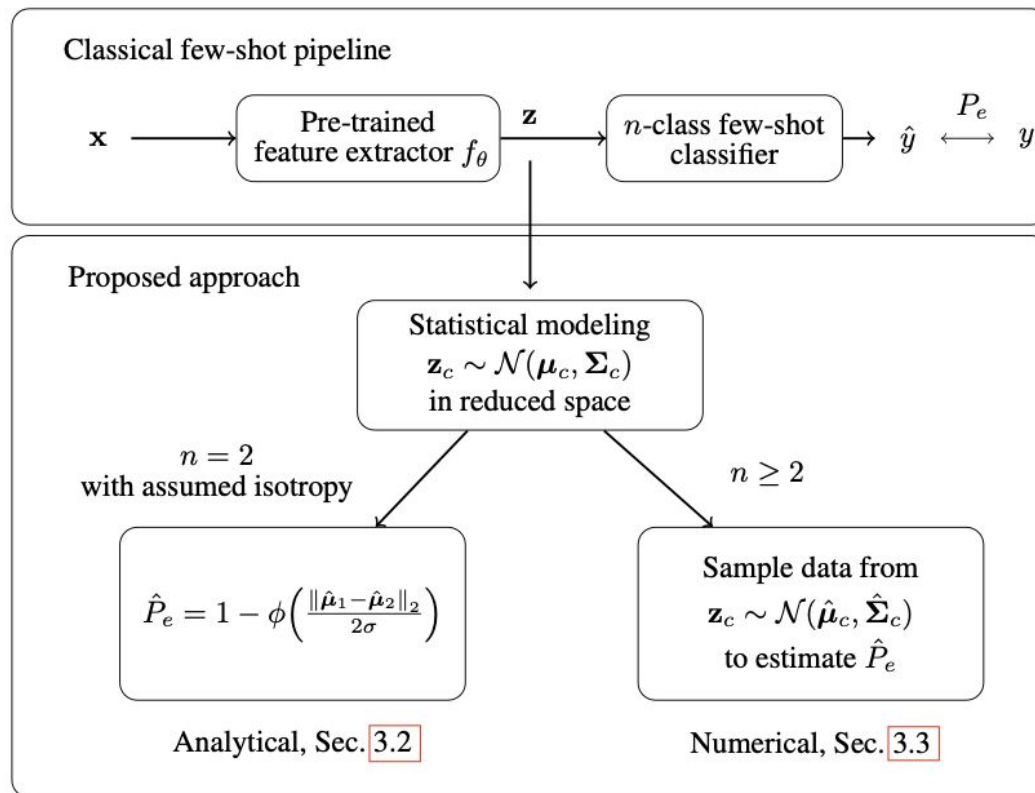
$$P_e = \sum_{c \in \mathcal{Y}} \int_{R \setminus R_c} p(\mathbf{z} \mid y = c) p(y = c) d\mathbf{z}$$

In practice intractable!
for complex boundaries or
multiclass.

Where R_c is the

$$R_c = \{\mathbf{z} \mid C(\mathbf{z}) = c\}$$

Analytical insight



Analytical insight

- Binary classification (class a and class b)
- Isotropic case.

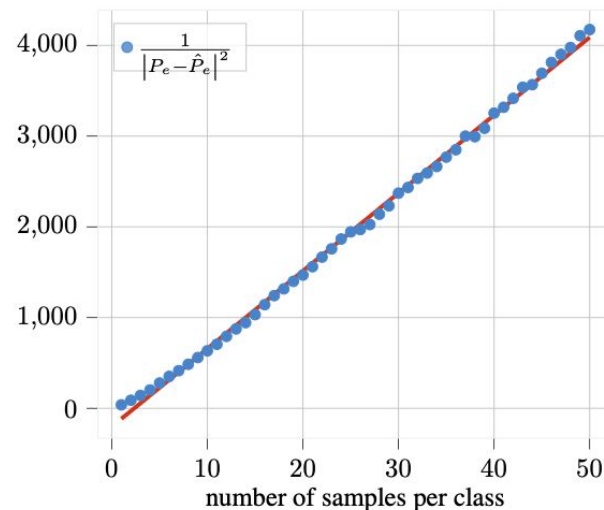
cumulative distribution function of the standardized Normal distribution

$$P_e = 1 - \phi\left(\frac{r}{2\sigma}\right), \quad r = \|\mu_a - \mu_b\|_2$$

Lemma 1 Given two classes represented by two univariate isotropic distributions $\mathcal{N}_a(\mu_a, \sigma^2)$ and $\mathcal{N}_b(\mu_b, \sigma^2)$ with shared and known standard deviations, the true probability of error P_e is bounded by:

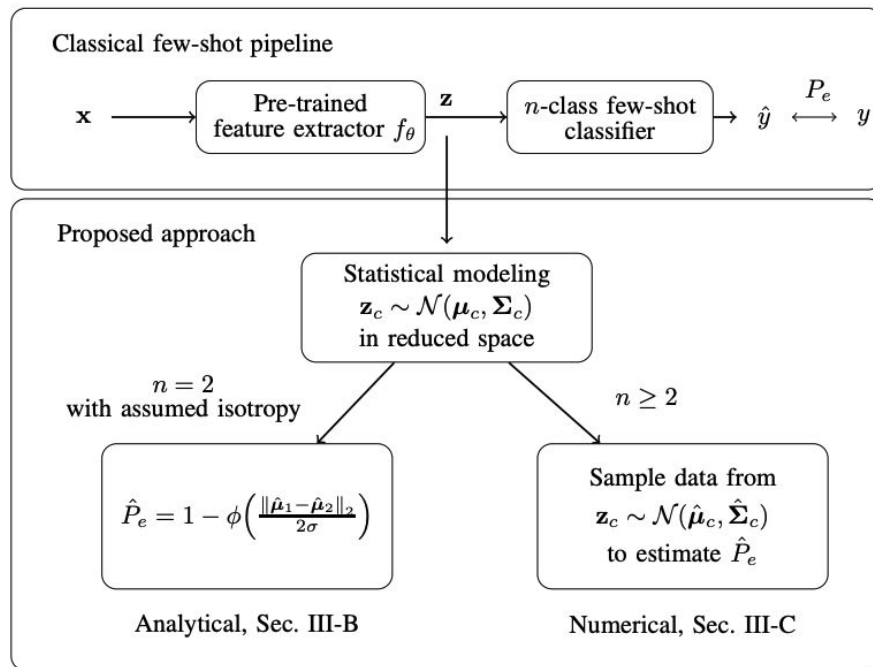
$$P\left(\left|P_e - \hat{P}_e\right| \leq \frac{\phi'(0)}{\sqrt{2k}} \left|\phi^{-1}\left(1 - \frac{\alpha}{2}\right)\right|\right) \geq 1 - \alpha, \quad (7)$$

with a probability $1 - \alpha$, where $\hat{P}_e$ is the probability error estimate from a sequence of k i.i.d random variables from the two distributions with empirical mean estimates $\hat{\mu}_a$ and $\hat{\mu}_b$, ϕ is the cumulative distribution function of $\mathcal{N}(0, 1)$, ϕ^{-1} its inverse and ϕ' its derivative, with $\phi'(0) = \frac{1}{\sqrt{2\pi}}$.



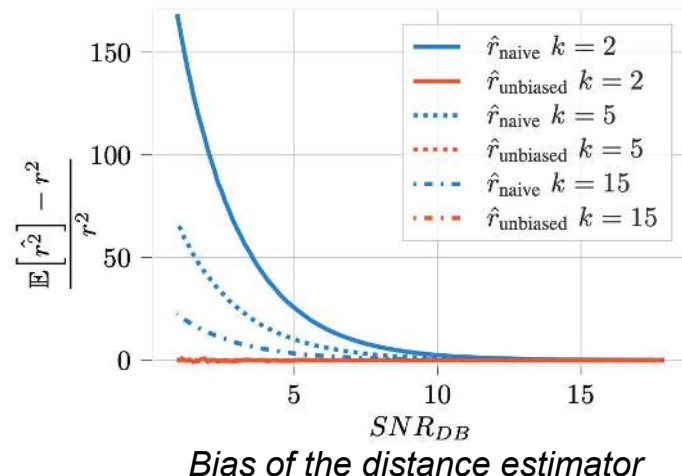
(a) 2-class, $R^2 = 0.9979$

A Statistical Model for Predicting Generalization in Few-Shot Classification

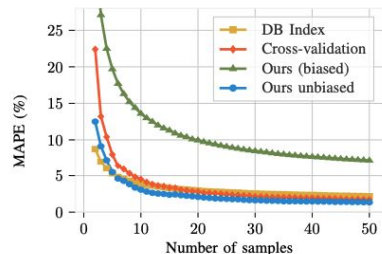


$$\hat{r} = \|\hat{\boldsymbol{\mu}}_a - \hat{\boldsymbol{\mu}}_b\|_2$$

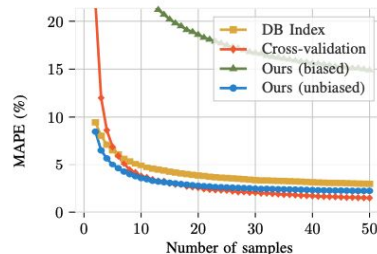
$$\mathbb{E}_{\mathbf{a} \sim p_a, \mathbf{b} \sim p_b}(\hat{r}^2) - r^2 = \frac{\text{Tr}(\boldsymbol{\Sigma}_a + \boldsymbol{\Sigma}_b)}{k}$$



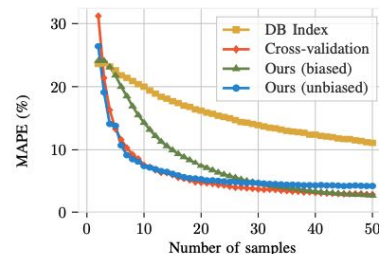
A Statistical Model for Predicting Generalization in Few-Shot Classification



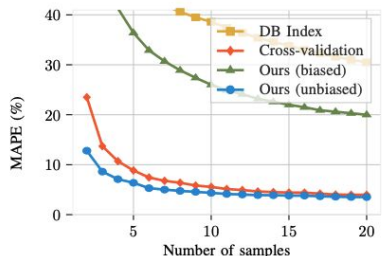
(a) Mini-ImageNet



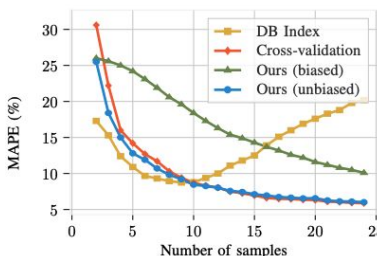
(b) Tiered-ImageNet



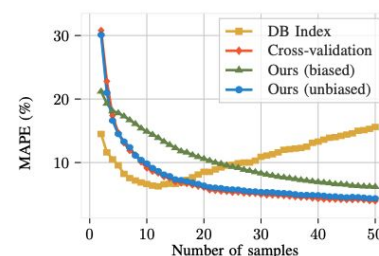
(c) ImageNet



(d) VGG-Flower



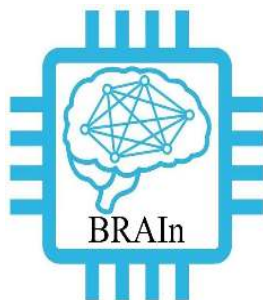
(e) CUB-200-2011



(f) Describable Textures (DTD)



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Performance on 10k 5-ways 1-shots FS problems from minilImageNet

